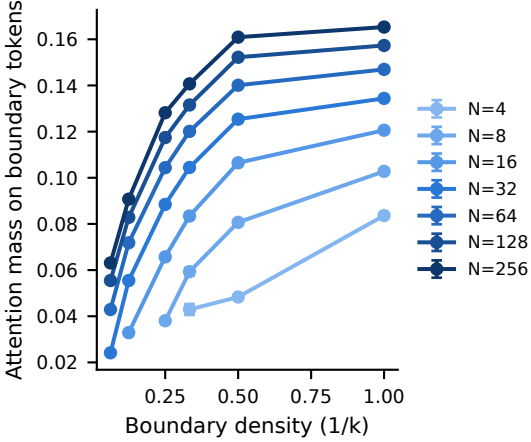
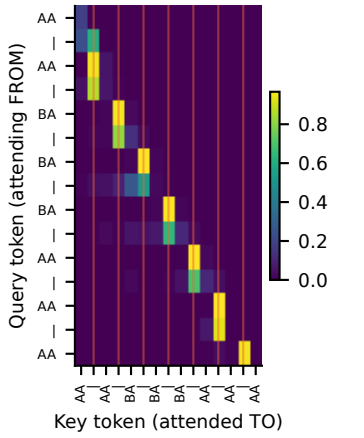
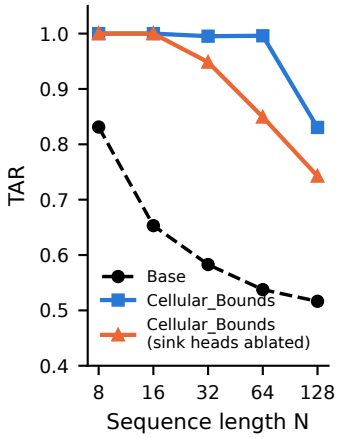
Boundary markers act as attention sinks, and that mechanism is causally necessary
(Qwen2.5-32B, cyclic_shift, alphabet=2, real tokenization)

(a) Boundary tokens absorb attention, scaling with density (alphabet=2)



(b) Attention heatmap, L21/H1 (orange lines = boundary tokens)



(c) Blocking sink heads erases a growing share of the benefit as N increases